

Master Theorem Worked Examples

Master Theorem Recap

The Master Theorem solves recurrences of the form:

$$T(n) = a \cdot T\left(\frac{n}{b}\right) + f(n)$$

where:

- $a \geq 1$: number of subproblems
- $b > 1$: input size shrink factor
- $f(n)$: cost of combining results

Let $E = \log_b a$. Compare $f(n)$ with n^E :

- **Case 1:** If $f(n) = O(n^{E-\epsilon}) \Rightarrow T(n) = \Theta(n^E)$
- **Case 2:** If $f(n) = \Theta(n^E) \Rightarrow T(n) = \Theta(n^E \log n)$
- **Case 3:** If $f(n) = \Omega(n^{E+\epsilon})$ and regularity condition holds, then $T(n) = \Theta(f(n))$

Example 1

Given: $T(n) = 4T(n/2) + n$

Solution:

- $a = 4, b = 2 \Rightarrow \log_2 4 = 2$
- $f(n) = n = O(n^{2-1})$
- Apply **Case 1** (polynomially smaller): $T(n) = \Theta(n^2)$

Example 2

Given: $T(n) = 2T(n/4) + \sqrt{n}$

Solution:

- $a = 2, b = 4 \Rightarrow \log_4 2 = 0.5$
 - $f(n) = \sqrt{n} = n^{0.5}$
 - $f(n) = \Theta(n^{\log_4 2})$
 - Apply **Case 2**: $T(n) = \Theta(n^{0.5} \log n)$
-

Example 3

Given: $T(n) = 8T(n/2) + n^3$

Solution:

- $a = 8, b = 2 \Rightarrow \log_2 8 = 3$
 - $f(n) = n^3 = \Theta(n^{\log_2 8})$
 - Apply **Case 2**: $T(n) = \Theta(n^3 \log n)$
-

Example 4

Given: $T(n) = 2T(n/2) + n^2$

Solution:

- $a = 2, b = 2 \Rightarrow \log_2 2 = 1$
- $f(n) = n^2 = \Omega(n^{1+\varepsilon})$ for $\varepsilon = 1$
- Check regularity: $2 \cdot (n/2)^2 = n^2/2 \leq c \cdot n^2$ for $c = 0.9$
- Apply **Case 3**: $T(n) = \Theta(n^2)$